

- Q.20 Give difference between two plate and three platemould.
- Q.21 Explain the use of CAD software in mould design and its advantages.
- Q.22 Describe procedure for detail model and prepare the bill of material.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)
- Q.23 Explain the different elements of injections mould and their functions.
- Q.24 What are the primary differences between thermoplastic and thermosetting plastics, and how do these differences impact their industrial applications?
- Q.25 Discuss the concept of allowances, gating system, cooling and ejection system of injection mould.

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2nd Year / Advance Diploma in Tool and Die Making **Subject : Tool Design-II (Plastic Moulds)**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What is the primary function of the mould housing and bases?
- a) Cooling and mould
 - b) Providing clamping force
 - c) Supporting the mould components
 - d) Ejection the moulded part
- Q.2 _____ material is self-fire extinguishing in nature.
- a) Polyethylene (PE)
 - b) Polypropylene (PP)
 - c) Poly Vinyl Chloride
 - d) Poly-acetal
- Q.3 In an injection moulding machine, what is the function of the clamping force?
- a) Plasticizing the material
 - b) Injecting the material into the mould
 - c) Holding the mould halves together
 - d) Cooling the moulded part

- Q.4 Which moulding process involves forcing a plastic material through a die to create continuous shapes?
- Blow moulding
 - Thermoforming
 - Extrusion moulding
 - Compression moulding
- Q.5 Which zone covers the 50% length of the injection moulding screw?
- Feed
 - Metering
 - Compression
 - Melting
- Q.6 Which classification of plastics includes materials that can be re-melted and re-moulded?
- Thermoplastic
 - Stabilizers
 - Pigments
 - None of above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Cooling channels is a mould help regulate the mould temperature, ensuring proper parts solidification. (True/False)
- Q.8 Give one application of electroplating and etching.
- Q.9 The primary function of the _____ system in plastic injection moulding is to control the flow of molten plastic.

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- Q.10 Define the term cavity in moulds.
- Q.11 Name two products made by Compression moulding.
- Q.12 CAD software is commonly used to design 3D models of plastic components for _____.

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Describe the safety measures that should be taken during mould maintenance.
- Q.14 What are the key properties of plastic materials?
- Q.15 Describe the cooling system in an injection mould and its importance.
- Q.16 Explain thermoforming process.
- Q.17 Explain classification of compression moulds.
- Q.18 Discuss the procedures for storing moulds and their significance.
- Q.19 Discuss the role simulation packages in plastic mouldings.

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